DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

CONTINOUS ASSESSMENT TEST I: EMT 3104: MECHATRONIC SYSTEM PROGRAMMING I

DATE: June 2025 **TIME: 1.5 HOURS**

---

a) Define what is meant by the term embedded system with regards to Mechatronic systems and give 2 suitable examples of such systems in daily life. **[2 marks]**

b) Using suitable well labelled diagrams, explain the 2 types of microcontroller interrupts and 3 ways that an interrupt can be triggered **[4 marks]**

c) Using a suitable diagram explain the main source of digital input noise such as push buttons and hardware and software debouncing techniques. **[6 marks]**

d) An ATmega32 series microcontroller contains several ports as general-purpose input and output (GPIO) ports. Discuss the 3 registers used to work with these ports and their functionality. **[6 marks]**

e) In micro-controller programming, binary, decimal and hexadecimal number systems are generally used.

i) Convert the following binary number to hexadecimal and decimal: **10101101_2** **[4 marks]**

ii) Compute the output of the C code below and the binary computations for each output **[8 marks]**

```
int main(){
    int a = 54, b = 23;
    int c = a & b;
    int d = a | b;
    int e = a ^ b;
    int f = (((~c)|d)&e);
    printf("c = %d, d = %d, e = %d, f = %d ", c, d, e, f);
    return(0);
}
```